

Ayush Rawat

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EDUCATION

Master of Science, Web and Data Science
University of Koblenz

04/2023 – 11/2026
Koblenz, Germany

Bachelor of Technology, Computer Science and Engineering
Guru Gobind Singh Indraprastha University

08/2017 – 07/2021
New Delhi, India

PROFESSIONAL EXPERIENCE

Internal Audit Data Analyst
Allianz Commercial

03/2026 – Present

- Developed and executed audit test cases to support internal audit engagements, enabling auditors to validate controls and identify risk areas across business processes.
- Built Power BI dashboards for continuous risk monitoring and converted recurring audit test cases into reusable dashboards, reducing repetitive manual testing and improving turnaround on future audits.
- Leveraged Allianz's internal LLM to build AI agents for the audit team, streamlining data analysis and standardizing documentation to improve efficiency and consistency.

Data Analysis/Data Science Intern
Samsung Semiconductors

08/2025 – 02/2026
Munich

- Developed and maintained Python based automation scripts and an internal Streamlit web application to streamline data entry and management in MS SQL Server database.
- Designed and implemented ETL pipelines, integrating SQL stored procedures and data transformation logic to process data from Excel and PDF sources, leveraging Samsung's in-house LLM system for advanced data parsing and automation.
- Created and optimised SQL Views with computed columns and developed Power BI dashboards to improve data accessibility, visualisation, and decision making.

Investment Steering Intern
Munich Re

09/2024 – 07/2025
Munich, Germany

- Engineered comprehensive PowerBI dashboards processing complex financial datasets, enabling data-driven investment decisions.
- Developed and optimized SQL queries to extract and analyze data from enterprise databases
- Conducted Exploratory Data Analysis (EDA) on assets from different legal entities to derive actionable insights and created dashboards in Power BI.

Data Analyst Working Student
Allianz Technology

10/2023 – 08/2024
Munich, Germany

- Created dynamic Microsoft PowerBI dashboards to aid the Talent Development department in making data-driven decisions.
- Utilized Microsoft Power Automate to design and implement efficient workflows, significantly reducing manual processing time and boosting operational efficiency.

Research Intern
Bundesinstitut für Berufsbildung

04/2023 – 09/2023
Bonn, Germany

- Developed and automated an ETL pipeline for job advertisement data using AWS S3, Apache Airflow, and PySpark, ensuring efficient data processing and storage in PostgreSQL.
- Created interactive dashboards and automated reports using Apache Superset, providing valuable insights for business users.

PERSONAL PROJECTS

Social Media AI Content Generator 

- Engineered a web application using **Streamlit** and **Google Colab** to automatically generate and optimize social media content through iterative refinement.
- Implemented a **Multi-Agent** workflow using **LangGraph** and **LangChain** for content generation, critique, and improvement.
- Integrated **Llama-3.3-70b-versatile** model with **Groq API** for high-quality text generation while managing computational efficiency.

Fine-Tuning Llama 3.2-3B-Instruct for Football Match History 

- Designed and fine-tuned **Llama 3.2-3B** LLM model using **Hugging Face Transformers** to generate football match results from 2003 to 2023 by team and season.
- Used **PEFT (Parameter-Efficient Fine-Tuning)** with **LoRA** in 4-bit precision to optimize memory and computational efficiency during training.
- Deployed the final model and tokenizer to the **Hugging Face Hub**, leveraging tools like **BitsAndBytes** for quantization and efficient storage.

SKILLS

Languages: Python, SQL, R, Scala, Linux, **Tools and Libraries:** Microsoft PowerBI, Power Automate, Git/Github, AWS, Azure, Excel, Tableau, Databricks, Azure Datawarehouse, SAP BW, Jupyter Notebook, Looker, Pandas, Numpy, PySpark, Tensorflow, PyTorch, Matplotlib, Keras, BeautifulSoup, Scikit-learn, Tensorflow, Dplyr, ggplot2, TidyR, Ollama, OpenAI, Huggingface, SFTTrainer, **Frameworks:** Spark, Databricks, TensorFlow, Hadoop, Airflow, R Shiny, Langchain, Langsmith, LlamaIndex, Huggingface, Langsmith, Streamlit